

Program 1: Interfaces.java

Problem Statement

Write a Java program to demonstrate multiple interface implementation and interface inheritance.

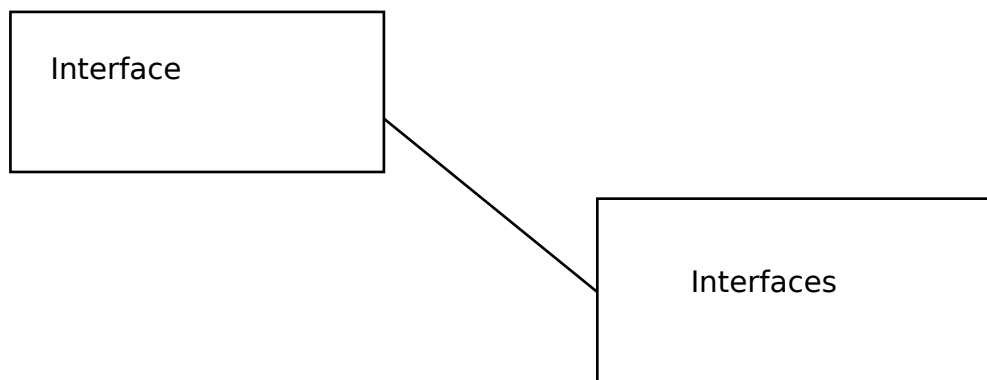
Description

Uses interfaces I2 and I3, where I3 extends I2. The Interfaces class implements both interfaces, displays interface constants, and calculates the sum of Y and Z.

Algorithm (10 Steps)

1. Start the program
2. Create interface I2
3. Create interface I3 extending I2
4. Declare Y and mult()
5. Create Interfaces class
6. Implement sample()
7. Implement mult()
8. Create object
9. Display values and call methods
10. Stop

UML Class Diagram



Java Code

```
package inheritance;

interface I2{
    int Z = 305;
```

```

    void sample();
}

interface I3 extends I2{
    int Y = 208;
    void mult();
}

public class Interfaces implements I2 ,I3 {
    @Override
    public void sample() {
        System.out.println("The application is about sample");
    }

    @Override
    public void mult() {
        System.out.println("sum of two numbers : " + (Y+Z));
    }

    public static void main(String[] args) {
        Interfaces i1 = new Interfaces();
        System.out.println(I3.Y);
        System.out.println(I2.Z);
        i1.mult();
        i1.sample();
    }
}

```

Expected Output

Console Output
208
305
sum of two numbers :513
The application is about sample

Program 2: ClasstoInterface.java

Problem Statement

Write a Java program to demonstrate a class implementing an interface.

Description

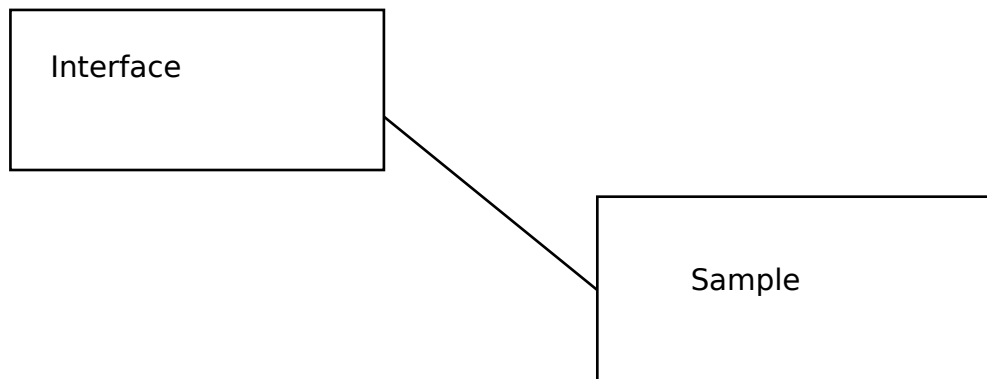
Sample implements interface I1, accesses interface constant y and class variable x, and calls both class and interface methods.

Algorithm (10 Steps)

1. Start the program
2. Create I1
3. Declare y
4. Declare text()
5. Create Sample

6. Declare x
7. Define add()
8. Override text()
9. Create object and display values
10. Stop

UML Class Diagram



Java Code

```
package inheritance;

interface I1{
    int y = 200;
    void text();
}
class Sample implements I1{
    int x = 100;
    void add() {
        System.out.println("This is a class ");
    }
    @Override
    public void text() {
        System.out.println("This is the interface");
    }
}

public class ClasstoInterface {
    public static void main(String[] args) {
        Sample sam = new Sample();
        System.out.println(I1.y);
        System.out.println(sam.x);
        sam.add();
        sam.text();
    }
}
```

Expected Output

Console Output

200

100

This is a class

This is the interface